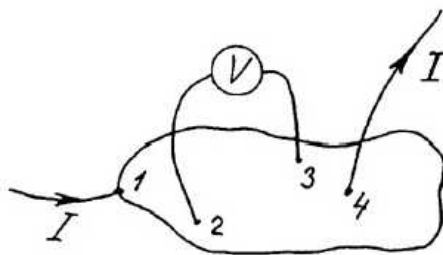


As the resistance of the contacts becomes comparable to sample resistance, it becomes necessary to separate the contacts conducting the current from the contacts between which the voltage is measured, in order not to measure contact resistance together with sample resistance. This is the rule when measuring resistivity of semiconducting materials, and also when measuring very small resistances of good conductors.

The separation of current and voltage contacts is fulfilled in the four-point method indicated in figure 2.

Figure 2:



Four-point measurement.

The resistivity is given by:

$$\rho = G \frac{V_{2-3}}{I_{1-4}} \quad (2)$$

where

V_{2-3} is the voltage between contacts 2 and 3,

I_{1-4} is the current through contacts 1 and 4.

From the reciprocal theorem of electromagnetic theory we conclude that if instead we pass the current I through contacts 2 and 3, we shall get between the contacts 1 and 4 the same voltage V as before between 2 and 3.

Thus we can extend equation (2) :

$$\rho = G \frac{V_{2-3}}{I_{1-4}} = G \frac{V_{1-4}}{I_{2-3}} \quad (3)$$